

EDUCATION

University of Alberta

Master of Science, Computing Science

Research Area: Reinforcement Learning and Search

Supervisor: Martin Mueller

GPA: 3.9/4

Edmonton, Canada

2021–2024

Amirkabir University of Technology - Tehran Polytechnic

Bachelor of Science, Computer Engineering

GPA: 17.92/20

Tehran, Iran

2016–2021

EXPERIENCE

Moving AI Lab

Research Assistant (Part-time)

- Topic: Evaluating reasoning capabilities of LLMs in planning.

Edmonton, Canada

02/2026 – present

MedWatch Technologies

Data Scientist

- Topic: Non-invasive blood glucose monitoring using AI

Florida, USA

01/2025 - present

Moving AI Lab

Research Assistant

- Topic: Designing Admissible Heuristics for Single-Agent Search Algorithms

Edmonton, Canada

04/2024 - 01/2025

University of Alberta

Graduate Research Fellow

- Topic: Improving sample efficiency of RL algorithms by Search and definition of Subgoals

Edmonton, Canada

09/2021 - 04/2024

University of Alberta

Graduate Teaching Assistant

- Courses: Intelligent Systems, Foundations of Computation I

Edmonton, Canada

09/2021 - 05/2022

Amirkabir University of Technology - Tehran Polytechnic

Teaching Assistant

- Courses: Algorithm Design, Operating Systems

Tehran, Iran

08/2020 - 04/2021

Institute for Research in Fundamental Sciences (IPM)

Research Internship

- Research area: solving games by AI

Tehran, Iran

07/2019 - 06/2020

PUBLICATIONS

1. **Ehsan Futuhi**, Nathan R. Sturtevant. “Learning Admissible Heuristics for A*: Theory and Practice”, *International Conference on Learning Representations (ICLR)*, 2026 (**28%** acceptance rate).
2. **Ehsan Futuhi**, Nathan R. Sturtevant. “A Parallel CPU-GPU Framework for Batching Heuristic Operations in Depth-First Heuristic Search”, *AAAI Conference on Artificial Intelligence (AAAI 2026)*, Accepted as Oral Presentation (Acceptance Rate: **17.6%**).
3. **Ehsan Futuhi**, Shayan Karimi, Chao Gao, Martin Müller. “ETGL-DDPG: A Deep Deterministic Policy Gradient Algorithm for Sparse Reward Continuous Control”, *Transactions on Machine Learning Research (TMLR)*, 2025.
4. **Ehsan Futuhi**. “Improving Deep Deterministic Policy Gradient for Sparse Reward and Goal-Conditioned Continuous Control”, *Master’s Thesis, University of Alberta*, 2024.
5. Kiarostami, Mohammad Sina, Mohammadreza Daneshvaramoli, Saleh Khalaj Monfared, Aku Visuri, Helia Karisani, Simo Hosio, Hamed Khashehchi, **Ehsan Futuhi**, Dara Rahmati, and Saeid Gorgin. “On using Monte- Carlo tree search to solve puzzles.”, *In 2021 7th International Conference on Computer Technology Applications*, pp. 18-26. 2021.
6. Sharareh Alipour, **Ehsan Futuhi**, Shayan Karimi. “On Distributed Algorithms For Minimum Dominating Set Problem, from theory to application.”, *preprint version available on arXiv*.
7. **Ehsan Futuhi**, Shayan Karimi. “Solving Nurikabe with Monte-Carlo Tree Search.”, *in 2020 5th International Conference on Computational Intelligence and Applications (ICCIA), IEEE, 2020*, pp. 33–38.

SELECTED PROJECTS

See full list of projects on my github

- Reasoning capabilities of LLMs in planning tasks (2026)
 - Developed novel environments for benchmarking LLM planning.
 - Introduced a specialized prompting strategy achieving 100% plan validity.
 - Fine-tuned state-of-the-art LLMs to establish robust performance baselines.
 - Conducted extensive experiments across diverse baseline models.
- Learning Admissible Heuristics (2025)
 - Developed a novel loss function for optimizing admissible heuristics in search.
 - Conducted sample complexity analysis for the learnability of admissible models.
 - Executed extensive experiments achieving admissible models for the Rubik’s Cube.
- ETGL-DDPG: Enhanced DDPG for sparse reward tasks (2025)
 - Design a new exploration algorithm.
 - Design a new replay buffer mechanism.
 - Extensive experiments on robotics benchmarks: Mujoco, and Meta-World.
- Batch IDA* (2024)
 - Implemented CPU and GPU parallelism to accelerate large-scale heuristic search.
 - Introduced a novel Batch IDA* variant capable of integrating ML-driven heuristics.
 - Leveraged generative models, including LLMs, to synthesize high-quality solution paths.
- Safe-MCTS (2023)
 - A MCTS approach for stochastic constrained MDPs.
 - A safety critic and policy prior attained by actor-critic algorithm.

- Policy Gradient (PG) Algorithms (2022)
 - Implement the state-of-the-art PG algorithms.
 - Evaluate exploration strategies in RL by their performance combined with PG algorithms.
- Analysing the game of Connect 4 (2021)
 - Solve the game by proof-number search.
 - Design MCTS and AlphaZero players.

SCHOLARSHIPS AND AWARDS

- B.Sc. Full scholarship for being in the top 700 of the National Public Universities Exam 2016

AREAS OF SPECIALIZATION

- **Reinforcement Learning & Robotics:** Deep RL, Policy Gradients, and sample efficiency with sparse-reward.
- **AI Planning & Search:** Heuristic Search and Monte Carlo Tree Search (MCTS).
- **Foundation Models:** LLM/VLM reasoning for planning, and RL fine-tuning.
- **Time Series Forecasting:** Blood glucose level prediction from non-invasive biosensor signals.

TECHNICAL SKILLS

- **Coding:** Python, Jax, Java, C/C++.
- **Deep Learning:** Pytorch/Libtorch, Tensorflow, PDDL.
- **Cloud & Infrastructure:** GCP, AWS, Azure , Docker, Kubernetes.

CONFERENCE ATTENDANCE

- International Conference on Learning Representations (ICLR) 2026
- Association for the Advancement of Artificial Intelligence (AAAI) 2026
- International Symposium on Combinatorial Search (SoCS) 2024
- Upper Bound AI Conference 2024
- Upper Bound AI Conference 2023

PEER REVIEWING EXPERIENCE

- International Conference on Learning Representations (ICLR) 2025-2026
- Association for the Advancement of Artificial Intelligence (AAAI) 2026
- International Conference on Automated Planning and Scheduling (ICAPS) 2025
- Uncertainty in Artificial Intelligence (UAI) 2024

LANGUAGES

- **English:** Fluent
- **Persian:** Native
- **Turkish:** Native